CHAPTER 1 :- The Wonderful World of Science

Topic 1: What is Science?

Definition Science is a way of thinking, observing and doing things to understand the world we live in and to uncover the secrets of the universe.

Concept Explanation Science is not just memorization; think of it as a big adventure where we ask questions, explore the world, and try to understand how things work. The absolute most important tool for this journey is "Curiosity". Science is everywhere around us, stretching from the tiny grains of sand to massive mountains, and from the depths of the ocean to the vastness of outer space. It functions like a giant and unending jigsaw puzzle. Every new discovery adds a piece to the puzzle, and because every new piece leads to more questions, there is no limit to what we can discover. Sometimes, new discoveries even force us to move puzzle pieces around, changing our understanding of the world.

Diagram

Real-Life Example Looking up at the night sky and wondering why the stars shine, or watching a flower bloom and wondering how it knows when to open, are perfect examples of science unraveling mysteries.

☑ **Quick Revision Checklist:**

- [ ] Science is a way of observing and understanding the universe.
- [ ] Curiosity is the driving force behind science.
- [ ] Science is like an unending jigsaw puzzle.
- [ ] New discoveries can change how we understand the world.

Topic 2: What Will We Explore? (The Scope of Science)

Definition The scope of science includes the exploration of planet Earth, its environments, diverse life forms, physical materials, and the celestial bodies beyond our planet.

Concept Explanation Our exploration starts with our home, planet Earth, which is the only planet we know that supports life. Science helps us understand the amazing variety of plants and animals surviving in different regions. We explore how a seed grows into a plant or how a caterpillar transforms into a beautiful butterfly. We also study vital non-living resources like food and water. For example, we observe that water freezes into ice when cooled and boils into steam when heated. Furthermore, we question what everyday objects—like paper, metal keys, plastic rulers, and rubber erasers—are made of and how to separate these materials. Finally, our questions don't stop at Earth; we can explore beyond the Sun, the Moon, and the millions of stars in the sky.

Diagram

Real-Life Example Jumping in a puddle when it rains and wondering why and how it rains, or figuring out how to separate different ingredients used to make tasty dishes across India's different cuisines.

☑ **Quick Revision Checklist:**

- [ ] Earth is the only known planet supporting life.
- [ ] Science studies plant and animal growth and survival.
- [ ] Science examines physical changes, like water turning to ice or steam.
- [ ] Science explores the materials that make up everyday objects.

- [ ] Scientific questions extend to the Sun, Moon, and stars.

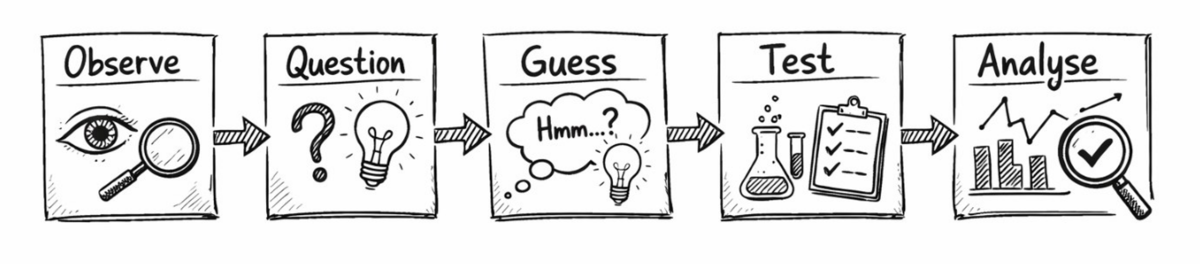
Topic 3: How Do We Find Answers? (The Scientific Method)

Definition The scientific method is a step-by-step process that helps us find answers to our questions through observation, guessing, testing, and analysis.

Concept Explanation Science is not about just knowing facts; it is about following a strict, logical process to solve problems. Here are the exact steps:

1. **Observe:** First, we observe something that we find interesting or do not understand.
2. **Question:** This makes us wonder and think of a question about it.
3. **Guess:** Then, we guess a possible answer to that question.
4. **Test:** We test this guess through experiments or more observations.
5. **Analyse:** We try to analyse the results to see if it actually answers our question.

Diagram



Real-Life Example If your pen stops writing, you ask, "Why did my pen stop?" (Question) . You guess the ink is finished (Guess). You open the pen to check the refill (Test). If the ink isn't finished, you make a new guess—maybe it dried up—and test that.

☑ **Quick Revision Checklist:**

- [ ] Science follows a step-by-step process to find answers.
- [ ] The process begins with observation and asking a question.
- [ ] We must guess an answer and then test it with experiments.
- [ ] Finally, we analyse the results to confirm our answer.

Topic 4: Working Like a Scientist

Definition Scientists are people who follow the scientific method to solve problems or to discover new things, but anyone who follows this method is working like a scientist.

Concept Explanation You do not need a laboratory to be a scientist. Whenever we try to ask questions and find out answers in our daily lives, we are all, in a way, scientists. Learning science develops our capabilities for finding solutions to bigger problems. To be a good scientist, the most important thing is to observe your surroundings keenly and ask "how and why?". Additionally, science is rarely a solo activity. Scientists across the world work together, often in large teams. So, if you are stuck, asking friends for help is a big part of the scientific process.

Diagram

Real-Life Example Someone cooking food wondering why the dal spilled out of the cooker (was there too much water?), or an electrician trying to find out if a broken light is due to the bulb or the switch, are both acting like scientists.

☑ **Quick Revision Checklist:**

- [ ] Anyone applying the scientific method is working like a scientist.
- [ ] Keen observation and asking "why" are the top scientific skills.
- [ ] Science is collaborative; scientists work together in teams.
- [ ] We unconsciously apply the scientific method to daily life problems.